SAKSHI GUPTA

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Professional Summary

B.Tech Electrical Engineering student targeting Software Engineer/SDE roles, with hands-on experience in Python backends, REST APIs, full-stack applications, RAG systems, and asynchronous job processing. Strong foundation in C++, Python, SQL, DSA, OOP, Operating Systems, DBMS, and Computer Networks; solved 350+ DSA problems.

Education

J.C. Bose University of Science and Technology, YMCA, Faridabad 2023–2027
B.Tech, Electrical Engineering

Experience

ExoHabitAI | Infosys Springboard Virtual Internship Feb 2026 – Apr 2026
Remote

- Developed a full-stack ML application using Next.js and Flask, integrating XGBoost for exoplanet habitability predictions and probability scores.
- Designed REST APIs for prediction and multi-planet ranking with JSON handling, input validation, error handling, and Joblib-based model loading.
- Integrated frontend and backend using asynchronous Fetch API calls, interactive result visualization, and preprocessing of NASA Exoplanet Archive data.

Tech Stack: Next.js, Tailwind CSS, Flask, Python, XGBoost, REST APIs

Campus Ambassador | Techfest, IIT Bombay Nov 2025 – Dec 2025

- Completed creativity- and ideation-focused technical challenges as part of the IIT Bombay Techfest Campus Ambassador Program.
- Secured AIR 323 in the Campus Ambassador Program while participating in technical outreach and event-related activities.

Projects

Distributed Job Processing System
Tech Stack: FastAPI, Python, Redis, PostgreSQL, Docker

- Built a distributed job-processing platform using FastAPI and Redis for asynchronous submission, queuing, and background execution.
- Designed a producer-worker architecture with job lifecycle tracking, retries, failure capture, result storage, and a Dead Letter Queue (DLQ).
- Integrated PostgreSQL for persistent job metadata, Docker for containerized infrastructure, and REST APIs with Swagger/OpenAPI for submission, monitoring, and result retrieval.

StudyLens | AI-Powered PDF Learning Assistant
Tech Stack: Python, FastAPI, PyMuPDF, LangChain, Sentence Transformers, RAG, LLM API

- Built a full-stack AI-powered PDF learning assistant providing context-aware question answering through a Retrieval-Augmented Generation (RAG) pipeline.
- Developed backend services for PDF extraction, document chunking, embeddings, semantic retrieval, and LLM response generation using FastAPI REST APIs.
- Integrated the frontend with the backend, extending the system toward page-specific Q&A, flashcards, personalized learning, quiz tracking, text-to-speech, semantic search, and multiple PDFs.

Technical Skills

Programming Languages: C++, Python, SQL, JavaScript, HTML/CSS
Backend & Web: FastAPI, Flask, REST APIs, PostgreSQL, Redis, Docker
Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks
AI/ML: Scikit-learn, PyTorch, TensorFlow, NLP, Sentence Transformers, RAG
Tools: Git, GitHub, VS Code, Jupyter, Google Colab

Certifications

Artificial Intelligence – Infosys Springboard | Principles of Generative AI – Infosys Springboard | Deep Learning – Infosys Springboard | Computer Vision | Intro to NLP

Leadership & Achievements

- **Session Head | Samarpan – The Technophillia**: Organized technical sessions on programming, DSA, and problem-solving for 25+ students; mentored peers in coding and algorithmic thinking and coordinated workshops and technical activities.
- Solved over 400 Questions on various coding platform such as **Leetcode, GeeksforGeeks, Codeforses.**
- Secured 6th rank in Quantafile Logo Innovation Jam 2024.